

Yang-Jun Joo, Ph.D.

Email: yangjunjoo.phd@gmail.com | Phone: +1 (407) 823-2841 | Location: Orlando, FL (Open to Remote / Relocation)
LinkedIn: linkedin.com/in/yang-jun-joo-a8012820a | GitHub: github.com/YangJunJoo | Google Scholar: scholar.google.com (Yang-Jun Joo)

EDUCATION

Seoul National University	Ph.D. in Civil & Environmental Engineering (Transportation Engineering) <i>Thesis: Generalized driving risk assessment in highway for autonomous vehicles using risk field approach (Advisor: Dr. Dong-Kyu Kim)</i>	Mar 2018 – Feb 2023
Seoul National University	B.S. in Civil & Environmental Engineering	Mar 2014 – Feb 2018

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar	University of Central Florida (Advisor: Dr. Mohamed Abdel-Aty)	Mar 2024 – Present
- Research autonomous driving safety evaluation and traffic safety analytics using advanced deep learning (including vision-language models) and conformal prediction. - Manage FDOT-sponsored corridors safety evaluation projects; designed large-scale data processing pipelines for high-resolution traffic signal logs (ATSPM) and vehicle telemetry.		
Postdoctoral Researcher	Seoul National University (Advisor: Dr. Dong-Kyu Kim)	Mar 2023 – Feb 2024
- Conducted smart mobility research, leading traffic safety analytics and data-driven travel choice modeling. - Acted as Principal Investigator (PI) on Sejong Science Fellowship, assessing passenger risk perception in driverless autonomous vehicle scenarios.		
Graduate Research Assistant	Seoul National University (Advisor: Dr. Dong-Kyu Kim)	Mar 2018 – Feb 2023
- Developed a generalized, real-time risk field-based safety evaluation metric for autonomous vehicles to proactive collision-avoidance. - Successfully secured competitive research funding (Global Ph.D. Fellowship) and published papers in top-tier journals (AMAR, AAP).		

SELECTED PROJECTS

Assessing Perceived Risk of Autonomous Vehicle Passengers	PI Sejong Science Fellowship (73.5M KRW)
Problem: Aligning AV behaviors with human risk expectations in near-miss/safety-critical events.	
Impact: Formulated surveys and regression models comparing AV vs. human driving passenger risk judgments; manuscript under 1st revision at IEEE T-ITS.	
Proactive Safety Opportunity Prediction using ATSPM Data	Researcher Florida Department of Transportation
Problem: Building real-time safety assessment pipelines using high-resolution signal controller event logs.	
Impact: Constructed scalable data processing workflows and dashboard-facing safety metrics to identify micro-level safety hotspots.	
NPTOI: Network Level Proactive Traffic Operations Indicator	Co-Researcher Active Research Project
Problem: Forecasting traffic states and network-level system performance under sparse connected vehicle telemetry.	
Impact: Developed spatiotemporal transformer models (MMCAformer) with cross-attention fusion for corridor speed predictions; manuscripts under review at IEEE T-ITS.	

TECHNICAL SKILLS

Data & Programming:	Python (pandas, NumPy, scikit-learn, PyTorch), R, SQL, large-scale mobility data workflows (ATSPM), statistics & experimentation.
Machine Learning & AI:	Conformal prediction (uncertainty calibration), Vision-Language Models (VLM) evaluation, generative AI (GANs for class imbalance), model validation.
Domain Expertise:	Autonomous driving safety evaluation, surrogate safety measures (SSMs), traffic safety & risk analytics, smart city infrastructure integration.